Evidence-based digital design: Utilizing MaxDiff findings to guide the development of financial interventions in bipolar disorder

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Introduction

Individuals with bipolar disorder (BD) are vulnerable to financial instability. Consequently, this population has shown interest in proactively mitigating illness-related financial harms. For example, 77% of individuals wanted an advance directive to name a financial proxy during acute illness states [1].

Modern financial data infrastructure can enable private, timely, highly-configurable financial interventions. Our research shows that individuals with BD are open to sharing financial data for use in digital tools for self-management [2, 3]. However, little is known regarding *what* tools might be preferred.

This study assesses individuals' relative preference among prototypical digital financial interventions.

Methods

Participants from the United States with BD (N=150) were recruited via Prolific in February 2026. We deployed a Maximum Difference Scaling (MaxDiff) survey [4] using Sawtooth Software. MaxDiff is a discrete choice design where participants identify the most and least important items in successive item subsets.

We assessed the relative importance of six hypothetical interventions targeting financial stability in BD. Three randomized prototypes were listed per task over eight tasks. Data analysis was performed on Sawtooth Software's Discover platform.

Drawing on repeated themes in our survey and interview data, we designed prototypes around the following concepts:

1. **adding purchase friction** (attention checks, cooling-off periods)
2. **third-party involvement** (collaborative budgeting, risky spending notifications)
3. **self-imposed spending limits**

Participants viewed a brief narrated video demonstration of each prototypes before completing MaxDiff tasks to ensure feature comprehension.

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Prototype Examples

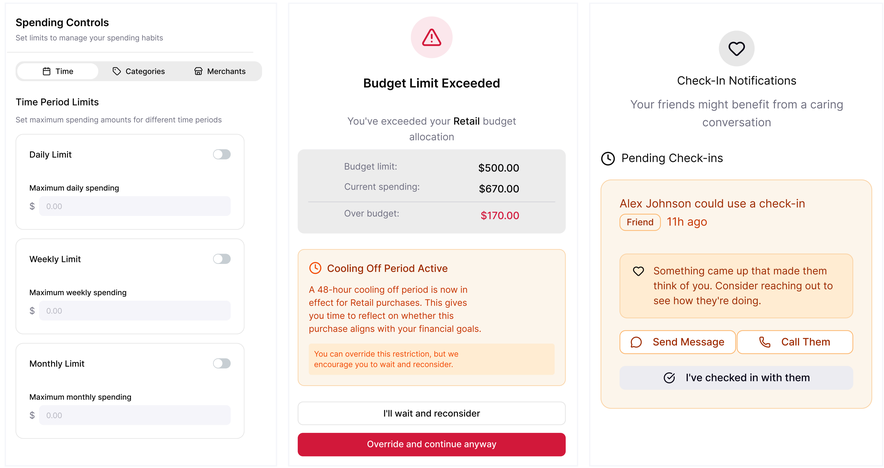


Figure 1. Among the six prototypes we included were **self-imposed spending limits** (left), **cooling-off periods combined with enforced budgeting** (middle), and **third-party overspending notifications** (right).

Preference for proposed digital financial intervention features (N=150)

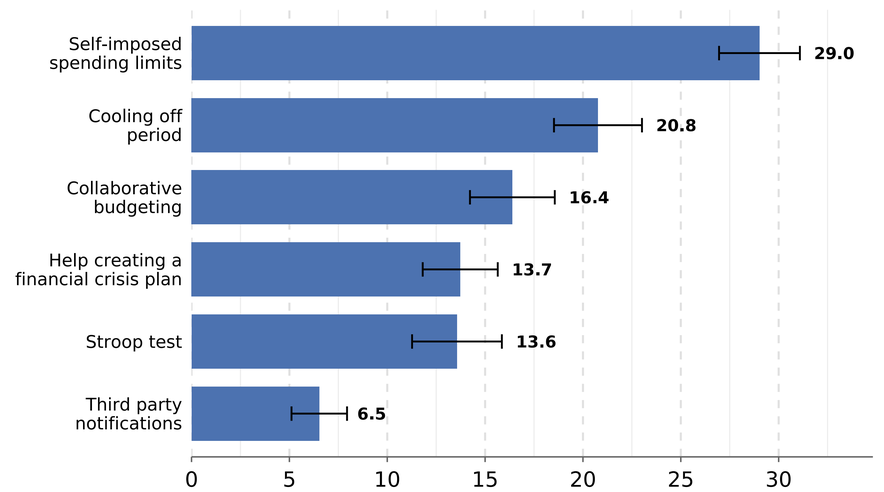


Figure 2. MaxDiff scores with 95% confidence intervals. Participants **significantly preferred self-imposed spending limits** (Score = 29.02, 95% CI [26.95,31.09]) **and cooling-off periods** (Score = 20.77, 95% CI [18.52,23.02]) over interpersonal interventions.

Demographics

Participants were predominantly female (71%) and white (71%). The sample median age was 40 years (IQR = 18). 34% were diagnosed with BD-I diagnosis, 37% with BD-II, 23% with BD-NOS, and 6% were not sure. All reported having been professionally diagnosed. **19% had previously declared bankruptcy and 59% had previously asked for help managing impulsive spending.**

Preferred features by diagnostic subtype

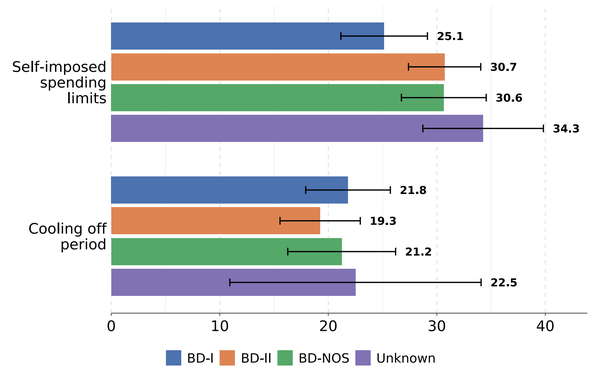


Figure 3. Relative preferences among diagnostic subtypes. We exclude cyclothymia as it was not represented in our sample.

Discussion

Our study indicates that individuals across a broad spectrum of diagnostic subtypes show a relative preference for self-managed financial interventions. These findings can help prioritize which intervention features best support financial stability in individuals living with bipolar disorder.

References

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